Muhammad Talha Atif

Software Engineer

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SUMMARY

Software Engineer with 1+ year of professional experience writing, reviewing, and debugging production code across Python, TypeScript, JavaScript, and Java. Practiced in tracing root causes through complex codebases, documenting decision points, and articulating technical reasoning clearly in written form. Experienced with Git-based workflows including pull requests, issue tracking, and code review across distributed teams. Strong foundation in data structures, algorithms, and multi-language problem solving through 500+ LeetCode problems solved and ICPC participation.

WORK EXPERIENCE

Software Engineer *Dec '25 – Present*
MicroTechX Remote

Built and debugged production backend services for an enterprise IAM platform, tracing issues across 8+ distributed Java microservices and documenting architectural decisions and debugging strategies.

- Diagnosed and resolved a Hibernate MultiBagFetchException by tracing ORM fetch strategy across entity relationships, documenting the root cause and fix for the team's reference.
- Debugged an $O(n^2)$ access-request lookup by analysing Java Collections usage, refactoring to O(n) map-based strategy and documenting the decision rationale clearly.
- Led zero-downtime OpenSearch migration reindexing 5M+ documents, documenting index template design and alias-switching strategy as team reference for future migrations.

Software Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Shipped and reviewed production features for a live social platform, tracing bugs across Node.js, TypeScript, and PostgreSQL layers with clear written documentation of fixes and reasoning.

- Traced a feed API bottleneck across database, query, and DTO layers, documented the 4-step fix strategy, and reduced response time from 3 seconds to 700ms.
- Identified a race condition causing notification duplication by analysing concurrent query patterns, documented the transactional fix and edge cases for the codebase.
- Reviewed and refactored 3,000+ line monolithic files into modular TypeScript components, documenting component boundaries and hook responsibilities for maintainability.

PROJECTS

Moodly | AI Experience Booking Platform | Demo

- Built Python FastAPI inference service integrating HuBERT and DeepFace models, debugging audio preprocessing edge cases across WAV format, sample rate, and channel conversions.
- Designed typed error schemas and structured JSON responses across 3 API endpoints, making failure modes explicit and traceable from the frontend integration layer.
- Documented event-driven architecture decisions for RabbitMQ pipeline design, capturing trade-offs between sync and async processing for team onboarding reference.

Memingo | Language Learning Platform | Demo

- Resolved leaderboard bottleneck by tracing recalculation logic, documenting the precomputed Redis cache strategy and reducing API response from 30 seconds to 90ms.
- Built real-time object detection module in Python, debugging classification pipeline output and structuring vocabulary responses as a single typed API contract.
- Integrated Gemini API chatbot with session-aware context, documenting prompt design decisions and structured output parsing logic for consistent cross-session behaviour.

SKILLS

Languages Python, TypeScript, JavaScript, Java, SQL, Kotlin
Backend Spring Boot, Node.js, Express.js, NestJS, FastAPI, REST APIs, gRPC
Frontend React.js, Next.js, React Native, Tailwind CSS, Shadcn UI
Databases PostgreSQL, MySQL, MongoDB, Redis, SQLite, Supabase, Firebase, OpenSearch
Git Workflows GitHub, pull requests, code review, issue tracking, branch strategy, CI/CD
Problem Solving 500+ LeetCode problems, ICPC participant, data structures, algorithms
Tools Postman, Swagger, Docker, Maven, VS Code, IntelliJ IDEA

AWARDS

Runner-up, App Development Hackathon (2025) *May '25*
Organized by Punjab University

Ranked, ICPC Asia Topi Online Programming Contest (2024) *Nov '24*
Organized by International Collegiate Programming Contests

Participant, CALICO Berkeley Speed Coding Competition (2024) *Dec '24*
University of California, Berkeley

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology – UMT *Oct '21 – Aug '25*